Evaluating LLM Reliability for Coding Adversarial Machine Learning Research Papers

***Abstract*—Meta-research on security and machine learning often requires systematic coding of large paper corpora, a task that has remained manual and slow. Large language models (LLMs) may help automate annotation, but their reliability for coding academic security papers is unknown. We evaluate LLM reliability for coding adversarial ML research by comparing four models (GPT-4o, GPT-5.2, Gemini-3, Claude Sonnet-4.5) to human expert coding on 71 papers and 9 categorical variables. All tested models reach nearly perfect intra-rater consistency (Fleiss' $\kappa = 0.97$–1.00), high inter-LLM agreement ($\kappa = 0.79$–0.85), and substantial agreement with the human coder ($\kappa = 0.64$–0.71). Reliability varies by category: structured variables like domain reach almost perfect agreement ($\kappa = 0.86$), while interpretive ones like attack category are moderate ($\kappa = 0.55$). LLMs are usable as screening tools for structured meta-research in security, though interpretive judgment still benefits from human review.**

1. Introduction

The security and privacy research community increasingly recognizes the need for critical self-reflection on research practices [1], [2], [3]. From evaluating peer review processes [4] to assessing reproducibility [5], meta-research enables evidence-based improvements to how we conduct, evaluate, and sustain scientific work. Careful meta-analysis still requires someone to code papers: methodology, evaluation practices, artifacts, threat models. That does not scale; manual coding is costly and usually keeps studies to tens of papers.

LLMs have shown decent reliability on text annotation in social science [6], [7] and medical reviews [8], with some agreement levels close to human annotators [9]. Nobody has yet tested whether they can reliably code *security* papers, where threat models, evaluation methodology, and jargon are specific. Reliable LLM coding could support large-scale meta-analysis; if not, knowing where they fail clarifies when expert coding is still needed.

Adversarial ML is a good testbed: the field has known evaluation issues [10], [11], [12], [13] (overstated robustness, weak adaptive evaluation), critiqued in manual work [14] but never in a large systematic coding study. Reliable automated coding would make that kind of meta-analysis feasible.

Contribution. We run the first evaluation of LLM reliability for coding security papers and ask: (1) How stable is each LLM across repeated codings (intra-rater)? (2) How well do LLMs match human coders? (3) Which variables are reliable or brittle under automation?

We use four LLMs (GPT-4o, GPT-5.2, Google Gemini-3, Claude Sonnet-4.5) on 71 adversarial ML papers and 9 categorical variables: research type, domain, venue, artifacts, threat model, and evaluation. Each LLM coded all papers 3 times; we compute Fleiss' κ for intra-rater and Cohen's κ for inter-rater agreement [15], [16].

Results are encouraging. All four models achieve nearly perfect self-consistency ($\kappa = 0.97$–1.00), high inter-LLM agreement ($\kappa = 0.79$–0.85), and substantial agreement with the human coder ($\kappa = 0.64$–0.71), in the range often deemed acceptable [17]. By category, structured variables (e.g., domain $\kappa = 0.86$, threat model $\kappa = 0.75$) reach high agreement, while interpretive ones (e.g., attack category $\kappa = 0.55$) are moderate.

LLMs look useful as *screening* tools for structured variables and could cut human effort on a first pass; they still fall short of expert coding on hard methodological calls. We report reliability numbers and release codebooks, prompts, and scripts for reuse.

2. Related Work

Recent work demonstrates that LLMs can match or approach human annotator performance on text classification tasks in social science [6], [7], medical systematic reviews [8], and content moderation [9]. Gilardi et al. [6] find GPT-3.5 achieves human-level agreement ($\kappa \approx 0.7$) on tweet classification, with GPT-4 outperforming crowd workers. Ziems et al. [18] systematically compare LLMs to human annotators across 27 datasets, finding that LLMs excel on structured tasks but struggle with subjective judgments. Törnberg [7] had GPT-4 beat experts and crowd workers on political tweets, while Rathje et al. [9] got human-like correlations ($r = 0.59$–0.77) across 15 datasets and 12 languages ($n = 47{,}925$). For systematic reviews the picture is mixed: Khraisha et al. [19] got adjusted $\kappa = 0.65$ for GPT-4 on screening, while a scoping review of 37 studies [20] called LLMs "not yet ready" as sole review tools. Gartlehner et al. [8], on 9,341 extractions from 63 medical studies, found Claude-assisted extraction at 91% accuracy vs. 89% human-only. Known issues include model drift [21], prompt sensitivity [22], and extraction hallucinations. No prior work has evaluated LLM coding of *security* papers, where threat

models, evaluation methodology, and jargon are domain-specific.

Content analysis uses standard agreement metrics: Cohen's κ [23] handles two raters with chance correction, while Fleiss' κ [24] extends to multiple raters. Krippendorff's α [15] is flexible for any number of coders; common advice is $\alpha \geq 0.80$ for reliability [25]. Landis and Koch [17] label $\kappa = 0.61$–0.80 "substantial" and 0.81–1.0 "almost perfect." Neuendorf [16] suggests $\kappa \geq 0.80$ for published work. We use Fleiss' κ for intra-rater reliability and Cohen's κ for inter-rater, reporting bootstrap 95% confidence intervals. For threat-model variables (T1/T2), we compute subset κ excluding N/A to avoid prevalence inflation [26].

Adversarial ML has well-known evaluation problems. Carlini and Wagner [10] showed defensive distillation gave almost no robustness despite 99.5% claims. Athalye et al. [11] showed 7 of 9 non-certified ICLR 2018 defenses relied on "obfuscated gradients"; Tramèr et al. [12] broke 13 more from top venues. Arp et al. [13] reviewed 30 papers over 10 years and listed 10 pitfalls (data snooping, sampling bias, lab-only evaluation). These critiques say *what* is wrong but not *how common* each pattern is—large-scale coding could answer that. Prior meta-research in security has examined peer review [4], reproducibility [5], and artifact evaluation [3], but typically analyzes only tens of papers due to manual coding costs. Olszewski et al. [5] found no clear gain in artifact availability after artifact-evaluation committees appeared (2020) across 750 ML security papers. Reliable LLM coding could enable systematic reviews of hundreds of papers to quantify trends across entire subfields.

3. Methodology

We extract 71 adversarial ML papers from top security conferences (IEEE S&P, CCS, USENIX Security, NDSS), ML venues (ICML, ICLR, NeurIPS), and popular artefacts (CleverHans, RobustBench). The papers are chosen based on their impact and visibility in the adversarial ML literature; this artifact-based selection may overrepresent papers with clearer reporting and available code relative to a random sample. An arXiv ID and title identify each paper. We extract full text from each PDF with PyMuPDF (first 15 pages, max 50,000 characters). All raters saw the papers in the same order. For code availability variables (G5/G6), this PDF-only input restriction may disadvantage LLMs relative to the human coder, who could access external resources (GitHub, project pages).

Each paper was coded on 9 categorical variables grouped into three themes: research characteristics (G1–G6), threat protection properties specific to attack papers (T1–T2) and practical evaluation (Q1). These coding variables are standard and are commonly seen in the literature in adversarial ML reviews like [13], [27]; our framework was inspired in particular by [27]. Table 1 includes the complete variable list and allowed values. Some categories intentionally overlap (e.g., "Attack/Defense") to capture papers spanning multiple contributions, reflecting real-world ambiguity in the literature.

We use four LLMs and one human rater (Table 2). Based on their state-of-the-art performance in coding papers, we chose GPT-4o, GPT-5.2, Gemini-3 and Claude Sonnet-4.5 for coding our 71 papers. Each LLM was prompted separately through independent API calls with the same codebook, definition, allowed values, and a 0 temperature for consistency. We made 3 runs for each of the four models. A single human rater coded all 71 papers based on the same codebook.

Before computing agreement, we normalised all the values by trimming whitespace, hyphens, underscores, and removing `None`, `N/A`, `null`, and empty values, collapsing them to `n/a` so that formatting noise does not cause disagreement. We use two standard metrics [15], [16] to compute agreement. For models with 3 runs, we compute Fleiss' κ [24] over the three runs with items = paper $\times$ variable. This measures the model self-consistency across different runs. For the Human coder with only 1 run, intra-rater reliability cannot be computed and is marked as N/A. For each pair of raters, we compute Cohen's κ [23] on coding consensus. We concatenate all 9 columns into 639 labels per rater (71×9) for a global 5×5 matrix (diagonal = Fleiss' κ for multi-run models, off-diagonal = Cohen's κ). We also report mean κ per variable (G1–G6, T1–T2, Q1) across pairs to see which codes are stable. Interpretation follows the Landis and Koch [17] scale: $\kappa < 0.21$ (slight), 0.21–0.40 (fair), 0.41–0.60 (moderate), 0.61–0.80 (substantial), 0.81–1.00 (almost perfect).

4. Results

Figure 1 shows the full agreement matrix. Diagonal entries report intra-rater reliability (Fleiss' κ); off-diagonal entries report inter-rater agreement (Cohen's κ). Each κ is computed over all 639 labels (71 papers $\times$ 9 variables) for the given rater pair.

All four LLMs show nearly perfect self-consistency (intra-rater reliability): Gemini-3 achieves $\kappa = 1.000$ (identical outputs across 3 runs), Sonnet-4.5 reaches $\kappa = 0.994$, while GPT-4o and GPT-5.2 both reach $\kappa = 0.975$. Temperature-0 sampling produces highly deterministic behavior. For inter-rater reliability, all four LLMs show high inter-model agreement ($\kappa = 0.79$–0.85), suggesting they converge on similar interpretations. The best LLM pair is GPT-5.2 $\leftrightarrow$ Gemini-3 ($\kappa = 0.846$), while Sonnet-4.5 achieves excellent agreement with other LLMs ($\kappa = 0.79$–0.81). Critically, all LLMs achieve substantial agreement with the human coder: Gemini-3 ($\kappa = 0.707$), GPT-5.2 ($\kappa = 0.675$), GPT-4o ($\kappa = 0.654$), and Sonnet-4.5 ($\kappa = 0.643$)—all in the "substantial" range that Landis and Koch [17] consider acceptable for published work. Table 3 shows mean κ across all 10 inter-rater pairs, ranked by reliability; per-variable heatmaps appear in Appendix A.

Structured variables perform best: application domain (G3) achieves almost perfect agreement ($\kappa = 0.86$), while threat-model access (T1), gradient requirement (T2), code availability (G5), and release timing (G6) all reach substantial agreement ($\kappa = 0.70$–0.75). These have well-defined

TABLE 1. CODING SCHEMA: 9 CATEGORICAL VARIABLES

Variable	Description	Allowed Values
Group 1: Research Characteristics		
G1	Paper type	Attack, Defense, Evaluation, Both, Attack/Defense, Attack/Evaluation, Defense/Evaluation, Attack/Defense/Evaluation
G2	Attack category (if attack)	Evasion, Poisoning, Privacy, Defense, N/A
G3	Application domain	Vision, NLP, LLM, Audio, Malware, Tabular, Cross, Cross-domain
G4	Publication venue category	ML, Security, Journal, arXiv-only
G5	Code currently available	Yes, No
G6	Code release timing	At-pub, Post-pub, Never
Group 2: Threat Model (Attack Papers Only)		
T1	Adversary model access	White, Black, Gray, White/Black, N/A
T2	Gradient computation required	Yes, No, N/A
Group 3: Practical Evaluation		
Q1	Evaluation on real-world system	Yes, No, Partial, N/A

TABLE 2. RATERS AND RUN CONFIGURATION

Rater	Model/Source	Runs	Consensus
GPT-4o	OpenAI API	3	Majority vote
GPT-5.2	OpenAI API	3	Majority vote
Gemini-3	Google Gemini 3.0 Pro	3	Majority vote
Sonnet-4.5	claude-sonnet-4-5-20250929	3	Majority vote
Human	Domain expert	1	N/A

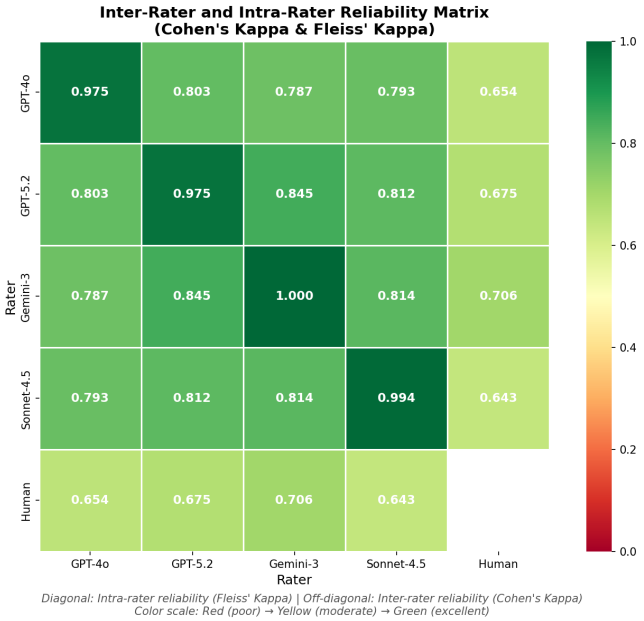

Figure 1. Global agreement matrix. Diagonal (dark green): intra-rater Fleiss' κ for all LLMs (3 runs each); N/A for the single-run Human coder. Off-diagonal: inter-rater Cohen's κ. Color scale: red (poor) $\rightarrow$ green (excellent).

categories with clear textual signals (e.g., "white-box," "black-box," "vision," "NLP"). Interpretive variables are harder: publication venue (G4), real-world evaluation (Q1), paper type (G1), and attack category (G2) achieve moderate

TABLE 3. PER-VARIABLE AGREEMENT (MEAN κ ACROSS ALL PAIRS WITH 95% CI)

Var	Description	Mean κ	95% CI	Interp.
G3	Application domain	.856	[.802, .901]	Almost perfect
T1	Threat model access	.748	[.674, .819]	Substantial
G5	Code available	.742	[.658, .821]	Substantial
T2	Gradient required	.736	[.644, .822]	Substantial
G6	Code release timing	.699	[.614, .780]	Substantial
Q1	Real-world evaluation	.614	[.508, .716]	Moderate
G4	Publication venue	.611	[.482, .734]	Moderate
G1	Paper type	.589	[.469, .706]	Moderate
G2	Attack category	.548	[.418, .674]	Moderate

agreement ($\kappa = 0.55$–0.61). These require judgment—is a paper that proposes an attack and then defends against it "Attack," "Defense," or "Both"? Is a poisoning attack also "privacy" if it extracts training data? Disagreements arise from ambiguous boundaries. The best pairs per variable reach substantial to almost perfect agreement: G3 (domain) achieves $\kappa = 0.87$ (GPT-5.2 $\leftrightarrow$ Sonnet-4.5), T2 (gradient) reaches $\kappa = 0.90$ (GPT-5.2 $\leftrightarrow$ Gemini-3), and G5 (code available) hits $\kappa = 0.89$ (GPT-5.2 $\leftrightarrow$ Gemini-3). Even G2 (attack category) achieves $\kappa = 0.79$ between GPT-5.2 and Sonnet-4.5.

One caveat concerns threat-model variables: while T1/T2 appear to reach substantial agreement in Table 3, subset κ excluding N/A reveals much lower LLM–human agreement ($\kappa = 0.18$–0.33). LLMs marked 56–70% of papers as N/A for T1/T2, while the human marked only 41–45%, indicating these variables are less reliable than global metrics suggest and benefit from N/A-driven prevalence inflation (see Limitations).

5. Discussion

Our results suggest that LLMs are viable for first-pass coding of structured metadata in security meta-research. For variables such as application domain ($\kappa = 0.86$), threat model ($\kappa = 0.75$), and code availability ($\kappa = 0.74$),

agreement exceeds inter-human reliability reported in other domains [18]. A practical workflow could employ LLMs to pre-code all papers, with human experts reviewing only disagreements or low-confidence cases.

Notably, all four tested models achieve substantial agreement with the human coder ($\kappa = 0.64$–0.71), with no single model dominating across all metrics. Gemini-3 demonstrates a slight advantage, achieving perfect self-consistency ($\kappa = 1.0$), the highest human agreement ($\kappa = 0.707$), and the lowest cost ($6.66 for 213 codings). Sonnet-4.5 also achieves nearly perfect self-consistency ($\kappa = 0.994$). The high inter-LLM agreement ($\kappa = 0.79$–0.85) suggests that these frontier models converge on similar interpretations of security papers, indicating that the choice of model may matter less than the quality of the codebook and prompt design.

However, interpretive variables remain more challenging. Attack category (G2, $\kappa = 0.55$) and paper type (G1, $\kappa = 0.59$) exhibit lower agreement than structured variables. The codebook permits overlapping labels (e.g., "Attack-/Defense"), and papers frequently span multiple categories. This pattern aligns with findings from social-science annotation [18], where tasks requiring inference about author intent consistently show lower LLM reliability. For such variables, LLM coding should be treated as a noisy signal that benefits from human adjudication.

These findings carry broader implications for adversarial ML meta-research. Our 71-paper corpus serves as a proof of concept demonstrating that reliable LLM coding could enable systematic studies of evaluation practices, artifact availability, and threat-model reporting across hundreds of papers—the kind of quantitative meta-analysis that manual coding has historically precluded. Researchers could, for instance, track whether artifact-evaluation policies [28] correlate with actual code availability over time, or examine whether vulnerabilities such as "obfuscated gradients" [11] persist in recent defenses. Our reliability numbers provide the first evidence that such large-scale studies are feasible for structured variables in security research.

6. Limitations

One of the major limitations of our work arises from resource constraints. First, we only use one expert coder due to the large number of papers and coding criteria. A single human coder may have bias, which can be mitigated by averaging or majority voting among three human raters. Prior work suggests human–human κ for similar tasks is 0.6–0.8 [15]; our LLM–human $\kappa = 0.71$ is strong, but this should be validated with multiple humans. Second, we were able to test only a few state-of-the-art models due to budget limitations for API calls. We would need to test in future with other models which show promise: Opus 4.6, o3 reasoning. As these models keep evolving, future models might show different results than models in early 2026. Third, we only used a single one-shot prompt and did not optimise. Per-model tuning or a few-shot examples could improve reliability. Fourth, the intra-rater consistency of models across three runs is not particularly significant and is largely a consequence of keeping the temperature to 0. Finally, the models might be influenced by their training data on the papers we included. The choice of top adversarial ML papers may not generalise to other security fields (systems security, cryptography).

7. Conclusion

We evaluated LLM reliability for coding adversarial ML papers on 9 categorical variables and found that all models achieve nearly perfect self-consistency ($\kappa = 0.97$–1.00) and substantial agreement with a human expert ($\kappa = 0.64$–0.71), with reliability varying by variable type: structured codes like domain reach almost perfect agreement ($\kappa = 0.86$), while interpretive codes like attack category are moderate ($\kappa = 0.55$–0.61). These results indicate that LLMs are usable as screening tools for structured meta-research in security, though they do not yet replace expert coding for nuanced methodological judgment. All supplementary materials including codebook, prompts, analysis scripts, raw CSV outputs, and per-model coding results are available at: https://github.com/actionproject-madhav/llm-paper-coding-reliability.

8. Ethics Considerations

We evaluate LLM coding of public academic papers. No human subjects; all papers are published and openly or institutionally accessible. No privacy, consent, or harm issues. The goal is to improve how the security community does meta-research.

9. LLM Usage Considerations

LLMs were both *evaluated* (as coders) and *used* (literature search, running experiments, analysis). We treat their coding output as data and validate it against human coding. LLMs helped with literature search, LaTeX formatting, running experiments, and debugging. Some experiment code is AI-generated with human verification. We release prompts, scripts, and raw outputs for replication.

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Appendix

Appendix: Supplementary Material and Coding Prompt

Figure 2 shows Cohen's κ for each variable across all rater pairs. Diagonal entries show intra-rater reliability (Fleiss' κ) for all four LLMs (each with 3 runs); the Human coder diagonal is N/A (single run). Variables G3 (domain), G5 (code available), T1 (threat model), and T2 (gradient) show consistent high agreement across most LLM pairs. Variables G1 (paper type), G2 (attack category), and Q1 (real-world evaluation) show moderate agreement, reflecting their interpretive nature.

All LLMs received identical system prompts with temperature set to 0.0 for deterministic outputs. Each paper was provided as extracted PDF text (first 15 pages, max 50,000 characters) with title and arXiv ID. Our corpus comprises 71 adversarial ML papers selected based on adoption in major security artifacts: CleverHans, IBM ART, Foolbox, AutoAttack, RobustBench, TextAttack, HarmBench, PyRIT, and MITRE ATLAS. Papers span 2013–2024 and include foundational attacks (e.g., FGSM, C&W), robustness benchmarks, certified defenses, and recent LLM jailbreaking work.

All supplementary materials are available at our GitHub repository: https://github.com/actionproject-madhav/llm-paper-coding-reliability, including raw CSV outputs from each model run (3 runs per LLM), human expert codings, the complete paper list with arXiv IDs and PDF links, analysis scripts for computing κ statistics, and API interaction code for each model.

Coding Prompt

```
You are coding academic papers for adversarial ML research. For each paper, assign values to exactly 9 variables. Use
     ONLY the allowed values listed below.

## CODING VARIABLES (STRICT VALUES)

### G1: Paper Type
ALLOWED: "Attack", "Defense", "Evaluation", "Both", "Attack/Defense", "Attack/Evaluation", "Defense/Evaluation", "Attack
     /Defense/Evaluation"

### G2: Threat Category
ALLOWED: "Evasion", "Poisoning", "Privacy", "Defense", "N/A"

### G3: Domain
ALLOWED: "Vision", "NLP", "LLM", "Audio", "Malware", "Tabular", "Cross", "Cross-domain"

### G4: Publication Venue
ALLOWED: "ML", "Security", "Journal", "arXiv-only"

### G5: Code Available
ALLOWED: "Yes", "No"

### G6: Code Release Timing
ALLOWED: "At-pub", "Post-pub", "Never"

### T1: Model Access Level (attacks only)
ALLOWED: "White", "Black", "Gray", "White/Black", "N/A"

### T2: Gradient Required (attacks only)
ALLOWED: "Yes", "No", "N/A"

### Q1: Real-World Evaluation
ALLOWED: "Yes", "No", "Partial", "N/A"

## OUTPUT FORMAT
Return ONLY valid JSON (no markdown):
{
  "G1": "<value>", "G2": "<value>",
  "G3": "<value>", "G4": "<value>",
  "G5": "<value>", "G6": "<value>",
  "T1": "<value>", "T2": "<value>",
  "Q1": "<value>",
  "reasoning": "Brief explanation"
}

CRITICAL: Use ONLY the allowed values. Do NOT create new values.
```

Figure 2. Per-variable agreement heatmaps showing pairwise κ for each coding variable (G1–G6: research characteristics; T1–T2: threat model; Q1: evaluation). Diagonal entries show intra-rater Fleiss' κ for all LLMs; Human is N/A (single run). Color scale ranges from red (poor) to green (excellent).